Project Assessment 5

Cross-Domain Replicated Analysis of Selection Perturbative Hyper-Heuristics with Explainable AI

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1 Introduction

This assessment addresses four feedback items from Assessment 4: insufficient experimental replication, an incorrect claim of comparability with published CHeSC results, inconsistent cross-technique SHAP magnitude comparisons, and the absence of explicit research questions to guide the analysis. The hyper-heuristic logic is unchanged; the improvements are in experimental methodology, statistical rigour, and explanation validity.

1.1 Research Questions

- **RQ1**: Within each domain, does any technique outperform the Simple Random baseline (SP-SR-LA) with statistical support?
- **RQ2**: Which technique performs best across domains?
- **RQ3**: Which state features drive acceptance within each technique, and are these explanations stable across seeds?
- **RQ4**: Do the features that drive acceptance differ between selection strategies, acceptance strategies, and search topologies?

1.2 Techniques

Technique	Topology	Selection	Acceptance
SP-SR-LA	Single-point	Simple Random	Late Acceptance, L=50
SP-CF-LA	Single-point	Choice Function	Late Acceptance, L=50
SP-CF-AILTA	Single-point	Choice Function	AILTA
MP-CF	Multi-point (pop=10)	Choice Function	Population replacement

Selection strategies. *Simple Random* selects a low-level heuristic uniformly at random each iteration, serving as a non-adaptive baseline. The *Modified Choice Function* (Drake, Özcan & Burke, 2012) scores each heuristic h as $\mathrm{CF}(h) = \varphi\, f_1[h] + \varphi\, f_2[prev][h] + (1-\varphi)\, f_3[h]$, where f_1 accumulates the improvement per unit CPU time of h applied alone (exponentially decayed), f_2 captures pair-sequence synergy between the previous heuristic and h, and f_3 is the elapsed time since h was last applied (favouring idle heuristics). The adaptive weight φ snaps to 0.99 on any

improving move and decays by 0.01 per non-improving step (floor 0.01), shifting the balance from exploitation to exploration as the search stagnates. The first n iterations (one per heuristic) apply each heuristic once in round-robin to initialise f_1.

Acceptance strategies. *Late Acceptance* (Burke & Bykov 2017) maintains a circular buffer of length L=50 storing past incumbent objectives. A candidate is accepted if it improves the current incumbent *or* is no worse than the entry L steps ago; the buffer is tightened monotonically (never relaxed). All slots are initialised to $+\infty$, so the first L iterations accept everything. *AILTA* is a simplified linear-decay variant of the Adaptive Iteration-Limited Threshold Accepting strategy used in AdapHH (Mısır & Özcan 2014). The full AILTA reheats the threshold after prolonged stagnation; our variant omits reheating and decays the threshold linearly: $\theta(t) = 0.02\,|f_0|\,(1-t/T)$, where f_0 is the starting objective. A candidate is accepted if its objective exceeds the incumbent by at most $\theta(t)$. At the final iteration, $\theta = 0$ and the strategy degenerates to improving-or-equal acceptance. The threshold is scaled to each domain's starting objective, avoiding the need to tune an absolute constant per domain.

Multi-point topology. MP-CF maintains a population of p=10 solutions. Each iteration, a binary tournament selects the *worst* individual as the replacement target, and a second tournament (excluding the target) selects the *best* as a potential crossover partner. A candidate replaces the target if and only if it is strictly better *and* the target is not the population-best (elitism guard). There is no separate acceptance module; the replacement rule serves as the acceptance criterion. A single shared Choice Function instance scores heuristics across all individuals.

Low-level heuristic sets. Each HyFlex domain exposes a fixed set of heuristics discovered at runtime via `getNumberOfHeuristics()`: SAT (11), BinPacking (8), FlowShop (15), TSP (13), VRP (10), PersonnelScheduling (12). These include mutation, ruin-and-recreate, local search, and crossover operators; the hyper-heuristic treats them as black boxes identified only by index and HyFlex type class.

2 Experimental Setup

2.1 Real HyFlex Domains

All experiments use the real CHeSC HyFlex jar files placed in `lib/hyflex/`. All 240 run manifests report `stub: false`. The six domains are SAT, BinPacking, FlowShop, TSP, VRP, and PersonnelScheduling.

Limitation: These results are not directly comparable to published CHeSC rankings. The CHeSC competition used a normalised 10-minute wall-clock time limit (scaled by a hardware benchmark), competition instances, and 31 independent runs per technique per domain. This study uses a per-domain iteration budget, instance 0 only, and 10 seeds. The real HyFlex domains are the same, but the experimental protocol differs.

2.2 Budget Policy

The budget is defined as a count of heuristic applications (calls to `applyHeuristic()`). All four techniques within a domain receive the same budget. MP-CF with population 10 receives the same total number of applications as the single-point techniques, not 10 times as many.

Per-domain budgets are set from convergence curves on a pilot run (Assessment 4, seed 42, 100,000 iterations for non-PS domains, 10,000 for PersonnelScheduling). For each domain, the budget is the smallest tier in {5,000; 10,000; 20,000; 50,000; 100,000} that exceeds the iteration at which all four techniques reach 95% of their total improvement.

Domain	Budget	95% convergence iteration
SAT	50,000	28,536
BinPacking	100,000	94,670
FlowShop	100,000	91,405
TSP	10,000	9,834
VRP	50,000	29,002
PersonnelScheduling	5,000	1,212

PersonnelScheduling's budget is naturally low because it converges rapidly. Profiling (Task 0) showed that 94.8% of PersonnelScheduling wall time is spent in HyFlex `LOCAL_SEARCH` calls (mean 484 ms per application), with instrumentation contributing only 1.4%. No optimisation of our code can reduce this cost.

2.3 Replication

Each cell (technique × domain) is run with 10 independent seeds: 42, 123, 456, 789, 1024, 2048, 3141, 4242, 5555, 9999. Each run uses a fresh problem object, fresh selector state, and a fresh RNG seeded with the run's seed. Total: 240 runs, all completing successfully. Total wall time: 20 hours 14 minutes.

2.4 Improvement over Assessment 4

Aspect	Assessment 4	Assessment 5
Seeds per cell	1	10
Total runs	24	240
Statistical tests	None	Wilcoxon, Friedman, A12
Budget	Fixed 100,000 all domains	Per-domain from convergence
SHAP surrogates	Pooled across domains	Per-domain, per-seed
SHAP comparison	Raw magnitudes	Ranks and shares
CHeSC claim	Claimed comparability	Explicitly disclaimed
`feat_llh_count`	Included (domain label)	Excluded

3 Results: Within-Domain Performance (RQ1)

3.1 Summary Statistics

SAT (budget: 50,000)

Technique	Median	IQR	Mean	SD	Min	Max
SP-SR-LA	12.50	[9.50, 15.75]	12.50	3.78	6.0	19.0
SP-CF-LA	11.50	[8.50, 13.75]	12.20	4.18	7.0	20.0
SP-CF-AILTA	**4.00**	[4.00, 4.75]	4.30	1.16	3.0	7.0
MP-CF	44.00	[42.25, 49.00]	43.80	8.31	30.0	56.0

BinPacking (budget: 100,000)

Technique	Median	IQR	Mean	SD
SP-SR-LA	0.01140	[0.0110, 0.0118]	0.0113	0.0006
SP-CF-LA	0.01190	[0.0114, 0.0124]	0.0119	0.0007
SP-CF-AILTA	0.02080	[0.0191, 0.0215]	0.0205	0.0020
MP-CF	**0.01140**	[0.0112, 0.0115]	0.0114	0.0002

Note: SP-SR-LA and MP-CF are tied at median 0.0114. MP-CF is not significantly better than SP-SR-LA ($p = 0.557$; see §3.2), so this ranking is not meaningful.

FlowShop (budget: 100,000)

Technique	Median	IQR	Mean	SD
SP-SR-LA	6353.0	[6347, 6366]	6354.3	16.6
SP-CF-LA	6364.0	[6346, 6374]	6362.1	30.0
SP-CF-AILTA	**6313.5**	[6303, 6325]	6311.4	18.5
MP-CF	6369.0	[6364, 6390]	6375.5	18.3

TSP (budget: 10,000)

Technique	Median	IQR	Mean	SD
SP-SR-LA	51,039	[50,607, 52,188]	51,206	1,070
SP-CF-LA	51,558	[51,198, 52,633]	51,640	1,257
SP-CF-AILTA	51,204	[50,231, 51,567]	50,996	1,036
MP-CF	**49,826**	[49,743, 50,100]	50,010	634

VRP (budget: 50,000)

Technique	Median	IQR	Mean	SD
SP-SR-LA	6,133	[5,896, 6,218]	6,123	669
SP-CF-LA	7,132	[7,020, 7,164]	7,031	584
SP-CF-AILTA	5,215	[5,188, 5,234]	5,299	289
MP-CF	**5,161**	[5,148, 5,197]	5,177	47

PersonnelScheduling (budget: 5,000)

Technique	Median	IQR	Mean	SD
SP-SR-LA	3,334	[3,315, 3,340]	3,330	24.3
SP-CF-LA	3,336	[3,323, 3,355]	3,340	24.8
SP-CF-AILTA	**3,301**	[3,291, 3,311]	3,303	14.2
MP-CF	3,341	[3,336, 3,345]	3,340	7.0

3.2 Wilcoxon Signed-Rank Tests vs SP-SR-LA Baseline

All tests are two-sided, paired by seed, with Holm correction within each domain. $A_{12} < 0.5$ means the tested technique is better (lower objective) than SP-SR-LA.

Domain	Technique	p (corrected)	A_{12}	Significant?
SAT	SP-CF-LA	0.992	0.455	No
SAT	SP-CF-AILTA	**0.006**	**0.01**	**Yes**
SAT	MP-CF	**0.006**	1.00	**Yes** (worse)
BinPacking	SP-CF-LA	**0.006**	0.90	**Yes** (worse)
BinPacking	SP-CF-AILTA	**0.006**	1.00	**Yes** (worse)
BinPacking	MP-CF	0.557	0.45	No
FlowShop	SP-CF-LA	0.541	0.59	No
FlowShop	SP-CF-AILTA	**0.012**	**0.03**	**Yes**
FlowShop	MP-CF	0.059	0.815	No
TSP	SP-CF-LA	1.000	0.61	No
TSP	SP-CF-AILTA	1.000	0.45	No
TSP	MP-CF	0.059	0.18	No
VRP	SP-CF-LA	**0.020**	0.81	**Yes** (worse)
VRP	SP-CF-AILTA	**0.020**	**0.19**	**Yes**
VRP	MP-CF	**0.018**	**0.12**	**Yes**
PS	SP-CF-LA	0.537	0.61	No
PS	SP-CF-AILTA	**0.041**	**0.16**	**Yes**
PS	MP-CF	0.449	0.695	No

3.3 Answer to RQ1

SP-CF-AILTA significantly outperforms the SP-SR-LA baseline in 4 of 6 domains (SAT, FlowShop, VRP, PersonnelScheduling; all $p < 0.05$ after Holm correction, all $A_{12} < 0.20$). In BinPacking it is significantly worse. MP-CF significantly outperforms the baseline in VRP ($A_{12} = 0.12$) but is significantly worse in SAT. In BinPacking and TSP, MP-CF's median advantage is not significant. SP-CF-LA shows no significant improvement over the baseline in any domain and is significantly worse in BinPacking and VRP.

The Choice Function with AILTA acceptance (SP-CF-AILTA) is the only technique that consistently improves over Simple Random across most domains.

4 Results: Cross-Domain Ranking (RQ2)

4.1 Rank Table

Techniques are ranked by median within each domain (1 = best, i.e. lowest objective).

Domain	SP-SR-LA	SP-CF-LA	SP-CF-AILTA	MP-CF
SAT	3	2	**1**	4
BinPacking	1.5†	3	4	1.5†
FlowShop	2	3	**1**	4
TSP	2	4	3	**1**
VRP	3	4	2	**1**
PS	2	3	**1**	4
Mean rank	**2.25**	**3.17**	**2.00**	**2.58**

†Tied at median 0.0114 (mid-rank); difference not significant ($p = 0.557$).

4.2 Normalised Scores

Normalised to $[0, 1]$ per domain: (median − best) / (worst − best). Lower is better.

Technique	Mean normalised score
SP-CF-AILTA	**0.30**
SP-SR-LA	0.49
MP-CF	0.50
SP-CF-LA	0.67

4.3 Friedman Test

Friedman chi-squared = 2.80, $p = 0.424$. **Not significant.** There is no statistically significant difference between the four techniques across all six domains when treated as a group. This is expected with only 6 blocks (domains) and relatively close mean ranks (2.00 to 3.17).

4.4 Answer to RQ2

SP-CF-AILTA has the best mean rank (2.00) and best mean normalised score (0.30), ranking 1st in 3 of 6 domains (SAT, FlowShop, PS) and 2nd in VRP. MP-CF ranks 1st in TSP and VRP and ties for 1st in BinPacking. However, the Friedman test does not detect a significant overall difference ($p = 0.424$), so the ranking advantage of SP-CF-AILTA over the alternatives cannot be confirmed at the cross-domain level with 6 domains.

No technique dominates across all domains. SP-CF-AILTA and MP-CF each have complementary strengths, but MP-CF's advantage is significant only in VRP.

5 SHAP Analysis (RQ3 and RQ4)

The XAI analysis addresses the benchmark paper's question—*which heuristics are effective, and when?*—indirectly: surrogates model acceptance (whether a heuristic application's result was kept) rather than performance directly. The features that the surrogate relies on to predict acceptance reveal which aspects of search state the hyper-heuristic implicitly conditions on when deciding to keep or discard a candidate solution.

5.1 Surrogate Fidelity

Per-domain, per-technique, per-seed gradient boosting surrogates were trained. `feat_llh_count` was excluded because it is constant within a domain and acted as a domain label in Assessment 4's pooled surrogates.

Technique	Mean ROC-AUC	Min ROC-AUC	Flagged (< 0.6)	Notes
SP-SR-LA	0.52	0.49	**60/60 (100%)**	Near-random
SP-CF-LA	0.88	0.66	0/60	Adequate
SP-CF-AILTA	0.91	0.70	0/60	Good
MP-CF	0.89	0.65	0/60	Adequate

SP-SR-LA surrogates are uniformly flagged across all 6 domains and all 10 seeds (ROC-AUC 0.49–0.55, mean 0.52). No SHAP attributions are reported for SP-SR-LA.

Robustness check. Because a uniform AUC of ∼0.5 could indicate a logging fault rather than a genuine absence of structure, we verified three potential failure modes: (1) `feat_last_h*` columns are populated with real heuristic indices for SR runs—the distribution across indices is approximately uniform, as expected from random selection; (2) acceptance labels are not near-constant—rejection rates range from 26% to 40%, comparable to CF-based techniques; (3) `feat_iters_since_accept` is recorded before the acceptance decision (the state snapshot is taken before the heuristic is applied and the decision made), so there is no target leakage.

The trace schema reveals the root cause: `feat_last_h1` records the heuristic applied in the *previous* iteration, not the current one (the current heuristic is in the `heuristic_id` column, which is not a surrogate feature). We verified this by checking that `feat_last_h1` at row N equals `heuristic_id` at row $N-1$ (100% match across all traces).

Acceptance rates grouped by the *actual* heuristic applied this iteration (`heuristic_id`) are *not* flat. In BinPacking, local search heuristics are accepted at ∼100% (they never worsen the solution, all deltas ≤ 0). Crossover is also accepted at 100%, but for a different reason: in single-point search the incumbent is crossed with itself, producing an identical child (delta = 0 in every application). One mutation heuristic (h3) also never worsens the solution and is accepted at 100%, while the other two mutation heuristics are accepted at only 0.4–1.9%. The per-type rates are: local search 100%, crossover 100%, ruin-recreate 57%, mutation 34% (the high type average reflects h3's 100% rate pulling up the two low ones).

Under random selection, `feat_last_h1` is statistically independent of the current heuristic, so it carries no signal about the current acceptance outcome and the surrogate cannot learn from it. Under the Choice Function, consecutive choices are correlated: CF never repeats the same heuristic (repeat rate 0% vs 12.5% expected under uniform random), and the f_2 pair-sequence term rewards certain orderings. This temporal structure makes `feat_last_h1` informative about which heuristic is applied *now*, giving the surrogate a learnable signal.

We therefore state the finding as: *the SR/CF explainability gap arises because the lagged heuristic features are uninformative under random selection but informative under adaptive selection's sequential structure.*

5.2 Feature Importance (Ranks and Shares)

SHAP importance is reported as **shares** (mean |SHAP| / total mean |SHAP|) and **ranks** (1 = most important). These are comparable across techniques within a domain. Raw SHAP magnitudes are never compared across techniques.

SP-CF-LA — Heuristic identity features dominate

Domain	Rank 1 (share)	Rank 2 (share)	Rank 3 (share)
SAT	`mean_logret_50` (33%)	`last_h1` (21%)	`last_h3` (16%)
BinPacking	`last_h3` (27%)	`last_h2` (20%)	`last_class1` (15%)
FlowShop	`last_h1` (27%)	`last_h2` (15%)	`last_h3` (13%)
TSP	`iters_since_imp` (32%)	`progress` (31%)	`mean_logret_50` (9%)
VRP	`last_h1` (15%)	`last_h2` (13%)	`last_h3` (12%)
PS	`last_h1` (18%)	`last_h3` (18%)	`iters_since_imp` (14%)

Feature names are abbreviated: `last_h`k = `feat_last_h`k (index of the k-th most recent heuristic).

In 4 of 6 domains, the top features are heuristic identity indicators (`feat_last_h*`). Because these features record the *previous* iteration's heuristic (§5.1), they are informative under CF

but not under SR. The Choice Function's sequential structure—no consecutive repeats and f_2 pair-sequence rewards—creates temporal correlations that let the surrogate infer the current heuristic (and hence the likely acceptance outcome) from the previous one.

SP-CF-AILTA — Search dynamics features dominate

Domain	Rank 1 (share)	Rank 2 (share)	Rank 3 (share)
SAT	`progress` (33%)	`iters_since_imp` (31%)	`improve_rate_50` (10%)
BinPacking	`last_h2` (27%)	`last_h3` (20%)	`iters_since_acc` (9%)
FlowShop	`improve_rate_50` (50%)	`progress` (20%)	`mean_logret_50` (7%)
TSP	`iters_since_imp` (29%)	`progress` (29%)	`improve_rate_50` (10%)
VRP	`progress` (30%)	`mean_logret_50` (20%)	`iters_since_imp` (16%)
PS	`last_h2` (20%)	`improve_rate_50` (16%)	`last_h1` (14%)

In 4 of 6 domains, the top feature is a search-progress indicator (`feat_progress`, `feat_improve_rate_50`, or `feat_iters_since_improve`). AILTA's linearly decaying threshold creates a dependency on *where in the search* the decision occurs, shifting importance away from heuristic identity toward search dynamics.

MP-CF — Population and temporal features emerge

Domain	Rank 1 (share)	Rank 2 (share)	Rank 3 (share)
SAT	`progress` (26%)	`iters_since_acc` (21%)	`pop_diversity` (18%)
BinPacking	`progress` (25%)	`iters_since_imp` (16%)	`gap_to_best` (10%)
FlowShop	`improve_rate_50` (22%)	`mean_logret_50` (22%)	`pop_diversity` (16%)
TSP	`mean_logret_50` (36%)	`iters_since_imp` (17%)	`iters_since_acc` (17%)
VRP	`iters_since_acc` (26%)	`pop_target_rank` (22%)	`pop_diversity` (15%)
PS	`progress` (26%)	`iters_since_acc` (12%)	`iters_since_imp` (11%)

Population-specific features (`feat_pop_diversity`, `feat_pop_target_rank`) appear in the top 3 in 3 of 6 domains (SAT, FlowShop, VRP). In VRP, where MP-CF ranks 1st in performance, `feat_pop_target_rank` has the highest share (22%)—the surrogate learned that the rank of the candidate within the population is a strong predictor of acceptance. The lagged heuristic features (`feat_last_h*`) are consistently relegated to ranks 9–15 in MP-CF. In multi-point search, consecutive applications typically target different individuals, so the previous heuristic carries even less signal than in single-point CF.

5.3 Stability Analysis (Kendall's Tau)

To assess whether SHAP explanations are stable across seeds, we compute Kendall's τ rank correlation between all 45 pairs of per-seed feature rankings for each technique–domain cell.

Domain	SP-CF-LA	SP-CF-AILTA	MP-CF
SAT	0.76 (0.08)	0.59 (0.18)	0.57 (0.11)
BinPacking	0.38 (0.30)	0.83 (0.09)	0.39 (0.14)
FlowShop	0.75 (0.08)	0.65 (0.10)	0.42 (0.14)
TSP	0.71 (0.10)	0.72 (0.10)	0.42 (0.20)
VRP	0.51 (0.11)	0.27 (0.17)	0.68 (0.07)
PS	0.64 (0.14)	0.72 (0.09)	0.62 (0.10)

Values are mean τ (std). $\tau > 0.6$: moderate–strong agreement; $\tau > 0.8$: strong.

SP-CF-LA achieves the highest stability in SAT (0.76) and FlowShop (0.75) but is unstable in BinPacking (0.38). SP-CF-AILTA is most stable in BinPacking (0.83) and TSP (0.72) but unstable in VRP (0.27). MP-CF is most stable in VRP (0.68) and PS (0.62) but weaker in BinPacking and FlowShop. No single technique produces universally stable explanations across all domains.

5.4 Answer to RQ3

The features driving acceptance differ by technique and are moderately stable across seeds (τ ranges from 0.27 to 0.83 depending on technique–domain cell, with most cells above 0.5). SP-CF-LA's decisions are explained by lagged heuristic identity features (acting as a proxy for the current heuristic under CF's sequential structure), SP-CF-AILTA's by search-progress features, and MP-CF's by population-level features. These structural differences between techniques are consistent across domains, even where per-seed stability within a single domain is lower.

Connecting this to the benchmark question of *which heuristics are effective when*: the SHAP explanations show that *what the surrogate needs to know* differs by technique. For SP-CF-LA, heuristic identity is the key signal—CF's sequential structure makes the previous heuristic predictive of acceptance. For SP-CF-AILTA, the search stage matters most, indicating that effectiveness is time-dependent. For MP-CF, population-level context (diversity, rank) matters more than the individual heuristic applied.

5.5 Answer to RQ4

Selection strategy: All 60 SP-SR-LA surrogates have ROC-AUC ≈ 0.52, while the 180 CF-based surrogates achieve mean ROC-AUC 0.88–0.91 (min 0.65–0.70). The gap arises because `feat_last_h*` features record the *previous* iteration's heuristic. Under random selection these are uninformative noise; under CF's sequential structure they predict the current heuristic and hence the acceptance outcome (§5.1).

Acceptance strategy: Comparing SP-CF-LA (Late Acceptance) with SP-CF-AILTA (AILTA), the acceptance strategy shifts importance from heuristic identity to search dynamics. AILTA's linearly decaying threshold makes acceptance sensitive to *where in the search* the decision is made, causing `feat_progress` and `feat_improve_rate_50` to dominate.

Search topology: Comparing SP-CF-LA (single-point) with MP-CF (multi-point), the population topology introduces `feat_pop_diversity` and `feat_pop_target_rank` as important features while suppressing heuristic identity features. In VRP, where MP-CF achieves its strongest significant performance advantage, `feat_pop_target_rank` (22%) and `feat_pop_diversity` (15%) together account for 37% of total SHAP importance.

6 LIME Analysis

6.1 Method

LIME explanations complement SHAP by providing interpretable *conditions* (e.g. `feat_progress <= 0.25`) rather than continuous attributions. For each surrogate, 50 evenly-spaced instances across `feat_progress` are explained. Each explanation condition is labelled with a sign: positive (+1) means the condition pushes the surrogate's prediction toward acceptance, negative (−1) toward rejection. Conditions are grouped into 10 progress bands (0–9) to track how acceptance drivers change over the search.

A condition is classified as an **acceptance driver** if its positive_fraction > 0.5 (i.e. it more often pushes toward acceptance than rejection across all explained instances where it appears).

Note that LIME conditions involving heuristic indices (e.g. `3 < feat_last_h2 <= 6`) use numeric index ranges. HyFlex exposes each heuristic's type class via `getHeuristicsOfType()`. For example, in BinPacking the mapping is: h0,3,5 = mutation; h1,2 = ruin-recreate; h4,6 = local search; h7 = crossover. In PersonnelScheduling: h0–4 = local search; h5–7 = ruin-recreate; h8–10 = crossover; h11 = mutation. The `feat_last_class*` features, which record the HyFlex type directly, provide a more interpretable alternative where they appear in explanations.

6.2 LIME Acceptance Drivers

The following summarises the most informative acceptance drivers per technique, selected from conditions with high positive_fraction and substantial count.

SP-CF-LA — Heuristic-specific conditions

- `feat_last_h3 <= 2.00` (BinPacking: count=144, pos_frac=0.77, weight=0.098)
- `3.00 < feat_last_h2 <= 6.00` (BinPacking: count=132, pos_frac=0.92, weight=0.165)
- `feat_last_h1 > 8.00` (PS: count=127, pos_frac=1.00, weight=0.112)
- `feat_progress <= 0.25` (SAT: count=103, pos_frac=0.95, weight=0.038)

SP-CF-LA's acceptance conditions depend on the *previous* iteration's heuristic. Under CF, the previous heuristic predicts the current one (§5.1), matching the SHAP finding that `feat_last_h*` dominates.

SP-CF-AILTA — Search-stage conditions

- `feat_progress <= 0.25` (SAT: count=130, pos_frac=1.00, weight=0.263)
- `0.25 < feat_progress <= 0.50` (SAT: count=120, pos_frac=1.00, weight=0.240)
- `feat_iters_since_accept <= 0.00` (BinPacking: count=364, pos_frac=1.00, wt=0.129)
- `feat_improve_rate_50 > 0.10` (FlowShop: count=72, pos_frac=0.90, weight=0.029)

The surrogate predicts higher acceptance probability in early progress bands (progress ≤ 0.50, positive_fraction = 1.0 across multiple domains). This is consistent with AILTA's decaying threshold, which is mechanically more permissive early. However, early-search permissiveness is not unique to AILTA: SP-CF-LA also shows `feat_progress <= 0.25` as an acceptance driver in SAT (pos_frac = 0.95). Part of the effect may simply reflect that improvements are easier to find early in any search, regardless of acceptance strategy.

MP-CF — Population diversity conditions

- `feat_pop_diversity > 0.05` (SAT: count=53, pos_frac=0.94, weight=0.308)
- `feat_pop_diversity > 0.00` (VRP: count=110, pos_frac=1.00, weight=0.263)
- `feat_pop_diversity > 0.01` (PS: count=70, pos_frac=0.86, weight=0.282)

- `feat_gap_to_best > 0.04` (BinPacking: count=90, pos_frac=0.93, weight=0.191)
- `feat_pop_target_rank > 6.00` (VRP: count=26, pos_frac=1.00, weight=0.202)

The surrogate predicts acceptance when population diversity is non-zero (positive_fraction 0.86–1.00 with high weights). This is consistent with the replacement rule: a candidate replaces the worst individual only if strictly better, and diverse populations offer more weak targets for replacement. The causal direction is not established—high diversity may be a *consequence* of recent acceptances rather than a *cause*—but the surrogate reliably uses it as a predictor.

6.3 LIME Stability

LIME feature-importance stability is measured by Kendall's τ on per-seed LIME share rankings.

Domain	SP-CF-LA	SP-CF-AILTA	MP-CF
SAT	0.52 (0.12)	0.54 (0.13)	0.47 (0.12)
BinPacking	0.34 (0.23)	0.69 (0.14)	0.41 (0.15)
FlowShop	0.39 (0.20)	0.59 (0.11)	0.44 (0.12)
TSP	0.40 (0.16)	0.54 (0.12)	0.63 (0.09)
VRP	0.21 (0.20)	0.34 (0.15)	0.61 (0.11)
PS	0.36 (0.20)	0.61 (0.10)	0.49 (0.13)

LIME stability is systematically lower than SHAP stability (mean τ 0.21–0.69 vs 0.27–0.83). This is expected: LIME produces local linear approximations at individual instances, so it is more sensitive to sampling variation than SHAP's global TreeExplainer attributions.

6.4 SHAP vs LIME Agreement

Both methods identify the same structural patterns:

1. **SP-CF-LA**: Heuristic identity features are the primary drivers in both SHAP (top shares go to `feat_last_h*`) and LIME (top conditions are heuristic-specific ranges).
2. **SP-CF-AILTA**: Search-progress features (*progress*, *improve_rate_50*) dominate both SHAP and LIME (progress ≤ 0.50, pos. frac. $= 1.0$).
3. **MP-CF**: Population features (*pop_diversity*, *pop_target_rank*) are key in both SHAP and LIME (pos. frac. ≈ 1.0).

LIME adds a complementary insight that SHAP cannot provide: the *direction* and *threshold* at which features drive acceptance predictions. SHAP tells us `feat_progress` is important for SP-CF-AILTA; LIME tells us that `feat_progress <= 0.50` drives acceptance with weight 0.24–0.26.

7 Limitations

1. **Not comparable to CHeSC**: This study uses iteration budgets, instance 0, and 10 seeds. CHeSC used a normalised 10-minute time limit, competition instances, and 31 runs. The real HyFlex domains are the same, but the experimental protocol differs.
2. **Single instance**: All runs use instance 0. Performance on other instances may differ.
3. **Friedman test power**: With only 6 domains (blocks), the Friedman test has limited power to detect moderate differences in technique rankings.

4. **PersonnelScheduling budget**: The 5,000-application budget is set by convergence analysis but is substantially lower than other domains. This domain's results rest on fewer applications.

5. **Convergence relative to pilot cap**: BinPacking (94,670) and FlowShop (91,405) reach 95% convergence very close to the 100,000-iteration pilot cap. The pilot run may itself not have been long enough, so the true budget needed may exceed 100,000 and results for these domains may not reflect fully converged search.

6. **SP-SR-LA unexplainable**: The Simple Random baseline cannot be explained by SHAP or LIME because its surrogates fail to learn acceptance patterns (ROC-AUC ≈ 0.52). This is explained by the lagged feature encoding: `feat_last_h*` records the previous iteration's heuristic, which is uninformative under random selection (§5.1). This limits direct feature-importance comparison to the 3 CF-based techniques.

7. **LIME stability**: LIME explanations are less stable than SHAP across seeds (mean τ 0.21–0.69 vs 0.27–0.83), particularly for SP-CF-LA in VRP ($\tau = 0.21$). Condition-level findings should be interpreted as aggregate trends, not per-seed guarantees.

8. **Surrogates model acceptance, not performance**: The XAI analysis explains which features predict whether a candidate is *accepted*, not whether it is *good*. Acceptance is a necessary condition for performance improvement but is not sufficient—a technique can accept many poor moves. The link between acceptance drivers and technique effectiveness is indirect.

8 Conclusions

1. **Replication works**: 240 runs with 10 seeds per cell provide reliable variance estimates and enable paired statistical tests.

2. **SP-CF-AILTA is the strongest single technique**: It significantly outperforms the baseline in 4 of 6 domains, ranks 1st in 3 domains by median, and achieves the best mean rank (2.00) and normalised score (0.30).

3. **MP-CF shows complementary strengths**: MP-CF achieves the best median in TSP and VRP (significant only in VRP) and ties for 1st in BinPacking, suggesting that population-based search can help on some problem structures. However, its advantage over the baseline is statistically confirmed only in VRP.

4. **The SR/CF explainability gap is a feature-encoding effect**: All 60 SP-SR-LA surrogates are flagged (ROC-AUC < 0.6), while the 180 CF-based surrogates achieve mean ROC-AUC 0.88–0.91. `feat_last_h*` records the *previous* iteration's heuristic. Under random selection this is independent of the current move, so the surrogate has no signal. Under CF, consecutive choices are correlated (no repeats, f_2 pair rewards), making the lagged feature informative about the current acceptance outcome. Actual per-heuristic acceptance rates (grouped by the current `heuristic_id`) vary widely (e.g. 0.4–100% in BinPacking), but the surrogate features do not include the current heuristic directly.

5. **Each technique type has a distinct explanation signature**: SP-CF-LA is explained by lagged heuristic identity, acting as a proxy for the current heuristic under CF's sequential structure. SP-CF-AILTA is explained by search-progress features, and MP-CF by population-level features. Both SHAP and LIME converge on this finding.

6. **AILTA's decaying threshold shifts the surrogate's reliance to progress features**: The surrogate for SP-CF-AILTA consistently ranks `feat_progress` and `feat_improve_rate_50` above heuristic identity features. LIME shows the surrogate predicts higher

acceptance early in the search, consistent with the linearly decaying threshold. However, early-search permissiveness also appears under SP-CF-LA, so this pattern may partly reflect general search dynamics rather than AILTA specifically.

7. **Population diversity predicts acceptance for MP-CF**: The surrogate for MP-CF relies on `feat_pop_diversity` and `feat_pop_target_rank`, consistent with the replacement rule that targets the worst individual. The causal mechanism is not established, but the association is strong in the domains where population features reach the top 3 (SAT, FlowShop, VRP).
8. **Cross-domain differences are not significant**: The Friedman test ($p = 0.424$) does not detect an overall technique effect, likely due to the small number of domains.